Introduction to Probability

Probability, Sample Space, Events, Simple Events

Objectives

- Discuss what probability is.
- Explain the use of probability in the real world.
- Discuss what events and sample space are.
- Demonstrate how to compute probability.

What is Probability?

*How **likely** something is to happen.*

The **probability** of an event refers to the likelihood that the event will occur.

Tossing a Coin

When a coin is tossed, there are two possible outcomes:

- heads (H) or
- tails (T)

We say that the probability of the coin landing **H** is ½

And the probability of the coin landing **T** is ½

Throwing Dice

- When a single die is thrown, there are six possible outcomes: **1, 2, 3, 4, 5, 6**.
- The probability of any one of them is *1/***6**

Probability

In general:

$$\text{Probability of an event happening} = \frac{\text{Number of ways it can happen}}{\text{Total number of outcomes}}$$

Example: the chances of rolling a "4" with a die

Example: the chances of rolling a "4" with a die

Number of ways it can happen: 1 (there is only 1 face with a "4" on it)

Total number of outcomes: 6 (there are 6 faces altogether)

$$\text{So the probability} = \frac{1}{6}$$

Example:

- There are 5 marbles in a bag: 4 are blue, and 1 is red. What is the probability that a blue marble gets picked?

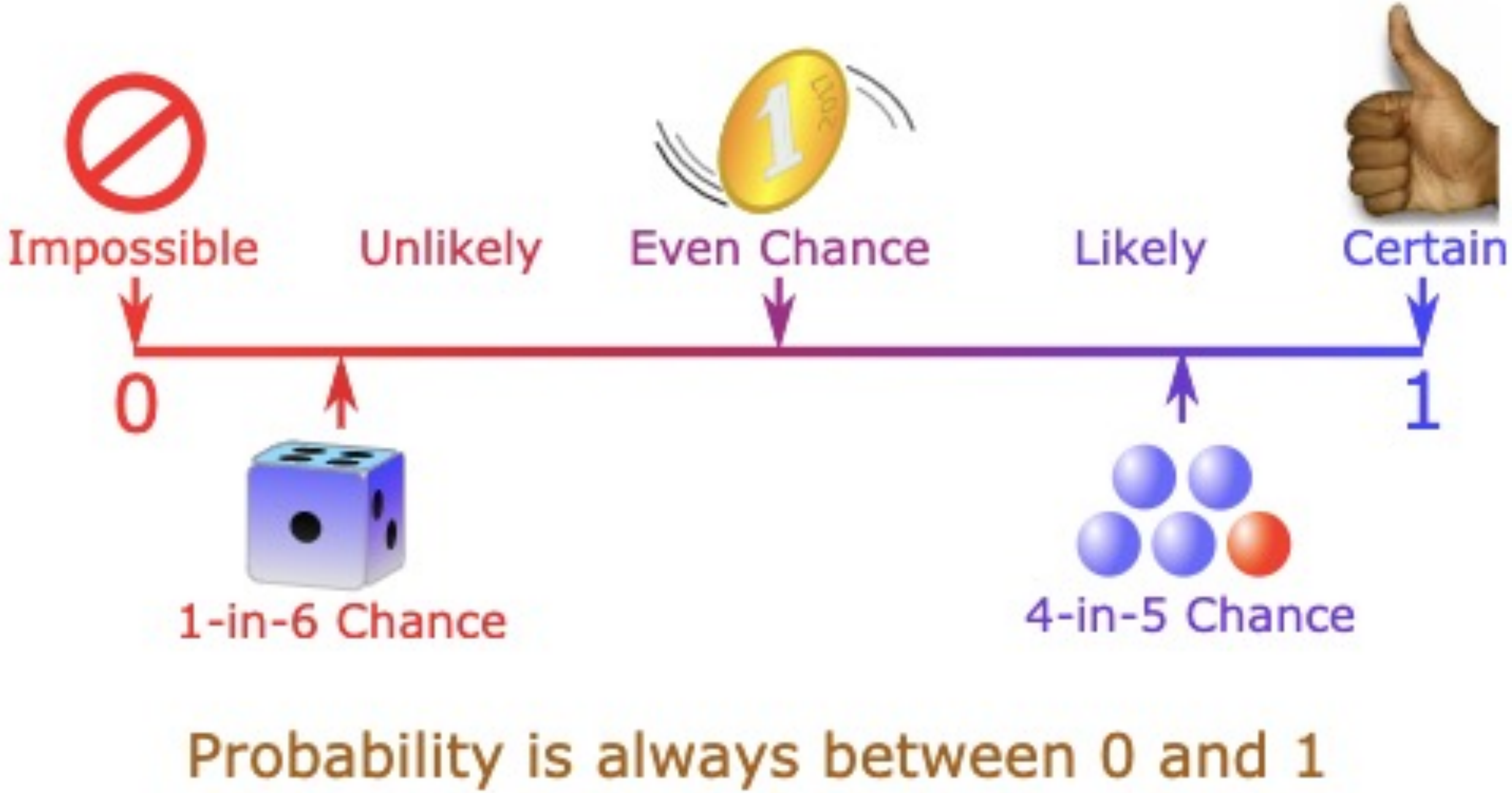
Number of ways it can happen: 4 (there are 4 blues)

Total number of outcomes: 5 (there are 5 marbles in total)

$$\text{So the probability} = \frac{4}{5} = 0.8$$

Probability Line

We can show probability on a Probability Line:



Probability Line

- Typically, probability is measured on a scale between 0 and 1. It can also be expressed in fractions, decimals or percentage form.
- If an event is ***impossible*** to occur, then the probability of that event is ***zero***.
- If an event is ***sure/certain*** to happen, then the probability of that event is ***one***.
- If an event has a 50-50, or an even chance of occurring, then the probability of that event is 0.5.
- An event that is ***unlikely*** to occur will have a probability between 0 and 0.5, while an event that is ***likely*** to occur will have a probability between 0.5 and 1.

Examples

- Scoring 120% in an examination is *impossible*.
- Tossing a coin and getting heads is an *even chance*.
- The sun will rise in the east is a *certain event*.
- Rolling a dice and getting a '2' on the top face is *unlikely*.
- A bag has 12 red balls, 5 green balls and 3 blue balls. Choosing a green ball is *unlikely*, but choosing a red ball is *more likely*.

EVENTS AND THE SAMPLE SPACE

- Data are obtained by observing either uncontrolled events in nature or controlled situations in a laboratory.
- We use the term experiment to describe either method of data collection.
- An ***experiment*** is the process by which an observation (or measurement) is obtained.

EVENTS AND THE SAMPLE SPACE

The observation or measurement generated by an experiment may or may not produce a numerical value.

Here are some examples of experiments:

- Recording a test grade
- Measuring daily rainfall
- Interviewing a householder to obtain his or her opinion on a no smoking city ordinance.

EVENTS AND THE SAMPLE SPACE

- Testing a printed circuit board to determine whether it is a defective product or an acceptable product
- Tossing a coin and observing the face that appears

When an experiment is performed, what we observe is an outcome called a ***simple event***, often denoted by the capital E with a subscript.

EVENTS AND THE SAMPLE SPACE

Definition: A ***simple event*** is the outcome that is observed on a single repetition of the experiment.

EVENTS AND THE SAMPLE SPACE

- Experiment: Toss a die and observe the number that appears on the upper face. List the simple events in the experiment.
- Solution When the die is tossed once, there are six possible outcomes. There are the simple events, listed below:
- Event E1: Observe a 1
- Event E2: Observe a 2
- Event E3: Observe a 3

Event E4: Observe a 4

Event E5: Observe a 5

Event E6: Observe a 6

EVENTS AND THE SAMPLE SPACE

- We can now define an ***event*** as a collection of simple events, often denoted by a capital letter.

Definition: An ***event*** is a collection of simple events.

EVENTS AND THE SAMPLE SPACE

We can define the events A and B for the die tossing experiment:

- A: Observe an odd number
- B: Observe a number less than 4

Since event A occurs if the upper face is 1, 3, or 5, it is a collection of three simple events and we write A ={E1, E3, E5}.

Similarly, the event B occurs if the upper face is 1, 2, or 3 and is defined as a collection or set of these three simple events: B={E1, E2, E3}.

EVENTS AND THE SAMPLE SPACE

- Note: *Sometimes when one event occurs, it means that another event cannot*.
- *Two events* are *mutually exclusive if*, when one event occurs, the others cannot, and vice versa.

EVENTS AND THE SAMPLE SPACE

- In the die-tossing experiment, *events A and B are not mutually exclusive*, because they have two outcomes in common—if the number on the upper face of the die is a 1 or a 3.
- In contrast, the *six simple events E1, E2, . . . , E6 form a set of all mutually exclusive outcomes of the experiment*.
- When the experiment is performed once, one and only one of these simple events can occur.

EVENTS AND THE SAMPLE SPACE

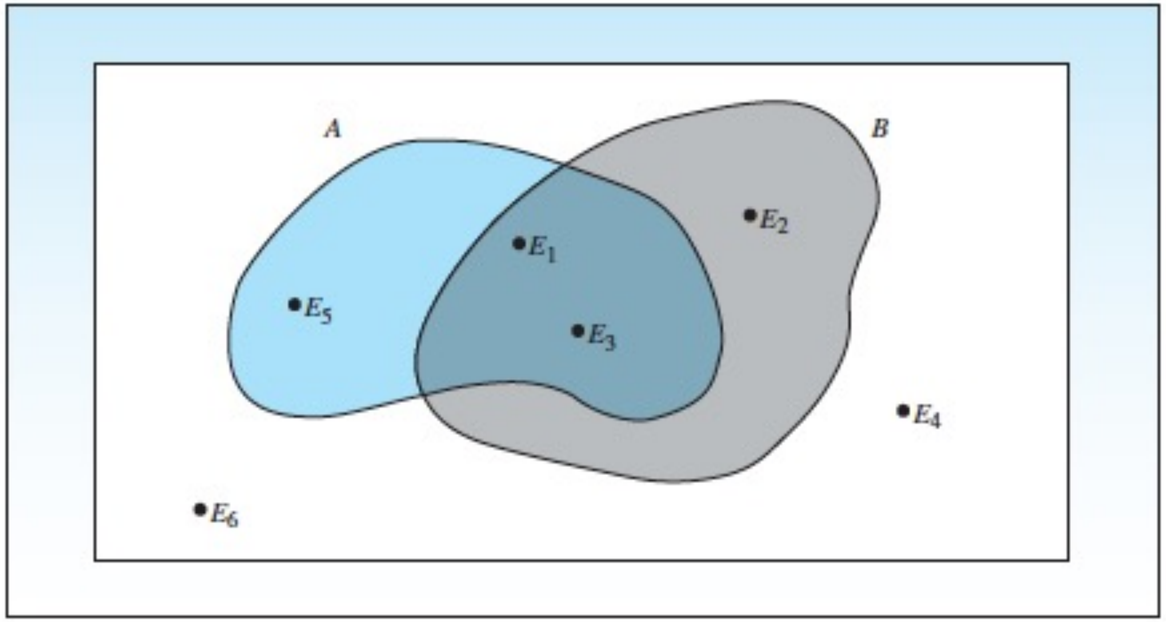
Definition: The set of all simple events is called the sample space, S.

EVENTS AND THE SAMPLE SPACE

- Sometimes it helps to visualize an experiment using a picture called a *Venn diagram.*

EVENTS AND THE SAMPLE SPACE

- The outer box represents the *sample space*, which contains all of the simple events, represented by labeled points.
- For the die-tossing experiment, the sample space is S ={E1, E2, E3, E4, E5, E6} or, more simply, S ={1, 2, 3, 4, 5, 6}.
- The events A ={1, 3, 5} and B ={1, 2, 3} are circled in the Venn diagram.



Experiment:

- Toss a single coin and observe the result. These are the simple events:

E1: Observe a head (H)

E2: Observe a tail (T)

- The *sample space is S ={E1, E2},* or, more simply, *S= {H, T}*.

Example: Experiment

Record a person's blood type. The four mutually exclusive possible outcomes are these simple events:

- E1: Blood type A
- E2: Blood type B
- E3: Blood type AB
- E4: Blood type O
- The *sample space is S ={E1, E2, E3, E4}*, or *S={A, B, AB, O}*.

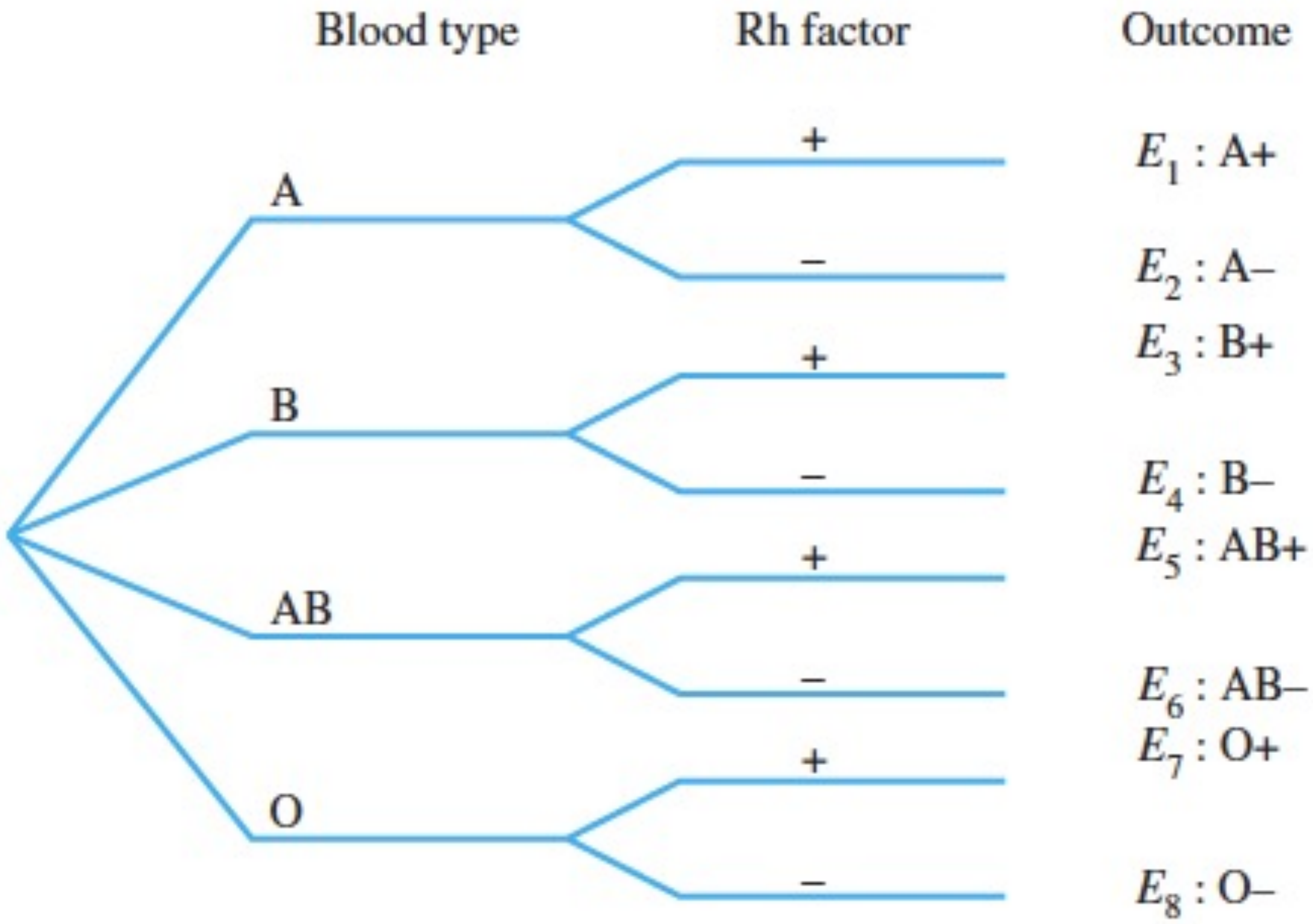
Tree Diagram

- Some experiments can be generated in stages, and the sample space can be displayed in a tree diagram.
- Each successive level of branching on the tree corresponds to a step required to generate the final outcome.

Example:

- A medical technician records a person’s blood type and Rh factor. List the simple events in the experiment.

Tree Diagram



Result

For each person, a two-stage procedure is needed to record the two variables of interest. The eight simple events in the tree diagram form the *sample space*,

S ={A+, A-, B+, B-, AB+, AB-, O+, O-}.

Probability Table

- An alternative way to display the simple events is to use a *probability table.*

Probability Table for Example

	Blood Type			
Rh Factor	A	B	AB	O
Negative	A−	B−	AB−	O−
Positive	A+	B+	AB+	O+

Notations and Rules

Notation for Probabilities

P denotes a probability.

A, B, and C denote specific events.

$P(A)$ denotes the probability of event A occurring.

Rule 1: Relative Frequency Approximation of Probability

Conduct (or observe) a procedure, and count the number of times that event A actually occurs. Based on these actual results, $P(A)$ is *estimated* as follows:

$$P(A) = \frac{\text{number of times } A \text{ occurred}}{\text{number of times the trial was repeated}}$$

Notations and Rules

Rule 2: Classical Approach to Probability (Requires Equally Likely Outcomes)

Assume that a given procedure has n different simple events and that *each of those simple events has an equal chance of occurring*. If event A can occur in s of these n ways, then

$$P(A) = \frac{\text{number of ways } A \text{ can occur}}{\text{number of different simple events}} = \frac{s}{n}$$

Notations and Rules

Rule 3: Subjective Probabilities

$P(A)$, the probability of event A, is *estimated* by using knowledge of the relevant circumstances.

Three Approaches to Finding Probability

(a)

(b)

(c)

Three Approaches to Finding Probability

- (a) Relative Frequency Approach (Rule 1): When trying to determine: P(tack lands point up), we must repeat the procedure of tossing the tack many times and then find the ratio of the number of times the tack lands with the point up to the number of tosses.

Three Approaches to Finding Probability

- (b) Classical Approach (Rule 2): When trying to determine P(2) with a balanced and fair die, each of the six faces has an equal chance of occurring.

$$P(2) = \frac{\text{number of ways 2 can occur}}{\text{number of simple events}} = \frac{1}{6}$$

Three Approaches to Finding Probability

- (c) Subjective Probability (Rule 3): When trying to estimate the probability of rain tomorrow, meteorologists use their expert knowledge of weather conditions to develop an estimate of the probability.

Three Approaches to Finding Probability

- It is very important to note that the classical approach *(Rule 2) requires equally likely outcomes*.
- If the outcomes are not equally likely, we must use the *relative frequency estimate* or we must rely on our *knowledge of the circumstances* to make an educated guess.

Three Approaches to Finding Probability

- When finding probabilities with the *relative frequency approach (Rule 1), we obtain an approximation instead of an exact value.*
- As the total number of observations increases, the corresponding approximations tend to get closer to the actual probability.
- This property is stated as a theorem commonly referred to as the ***law of large numbers***.

Law of Large Numbers

- As a procedure is repeated again and again, the relative frequency probability (from Rule 1) of an event tends to approach the actual probability.

Rule 1

- This is called ***Experimental Probability*** – probability derived from a series of trials, or experiments.
- The probabilities obtained from experimental probability may differ as we increase the number of trials.

Rule 2

- In theoretical probability, we expect the occurence of outcomes to be equally likely.
- When all outcomes are equally likely, then the theoretical probability of an event E is given by:

$$\text{Probability of E or P(E)} = \frac{\text{number of favourable outcomes to E}}{\text{total number of outcomes possible}} \quad \text{or } P(E) = \frac{n(E)}{n(S)}$$

Probability and Outcomes That Are Not Equally Likely

- One common mistake is to incorrectly assume that outcomes are equally likely just because we know nothing about the likelihood of each outcome.

EXAMPLE: Sinking a Free Throw

- Find the probability that NBA basketball player Reggie Miller makes a free throw after being fouled. At one point in his career, he made 5915 free throws in 6679 attempts (based on data from the NBA).

SOLUTION

- The sample space consists of two simple events: Miller makes the free throw or he does not. Because the sample space consists of events that are *not equally likely*, *we can't use the classical approach (Rule 2)*.
- We can use the relative frequency approach (Rule 1) with his past results. We get the following result:

$$P(\text{Miller makes free throw}) = \frac{5915}{6679} = 0.886$$

EXAMPLE: Genotypes

- As part of a study of the genotypes AA, Aa, aA, and aa, you write each individual genotype on an index card, then you shuffle the four cards and randomly select one of them.
- What is the probability that you select the genotype Aa?

SOLUTION

- Because the sample space (AA, Aa, aA, aa) in this case includes equally likely outcomes, we use the classical approach (Rule 2) to get

$$P(\mathrm{Aa}) = \frac{1}{4}$$

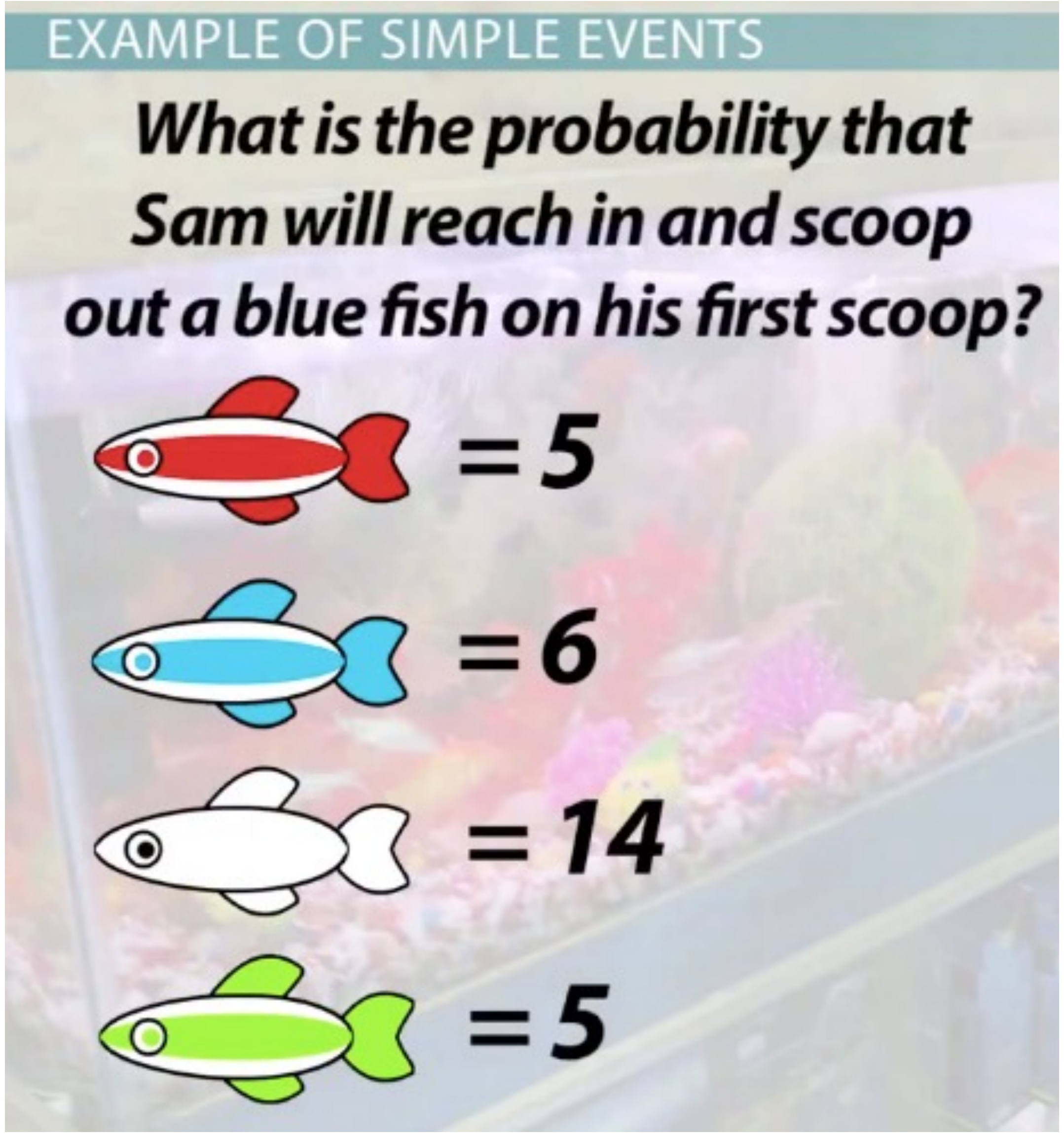
EXAMPLE: Crashing Meteorites

- What is the probability that your car will be hit by a meteorite this year?

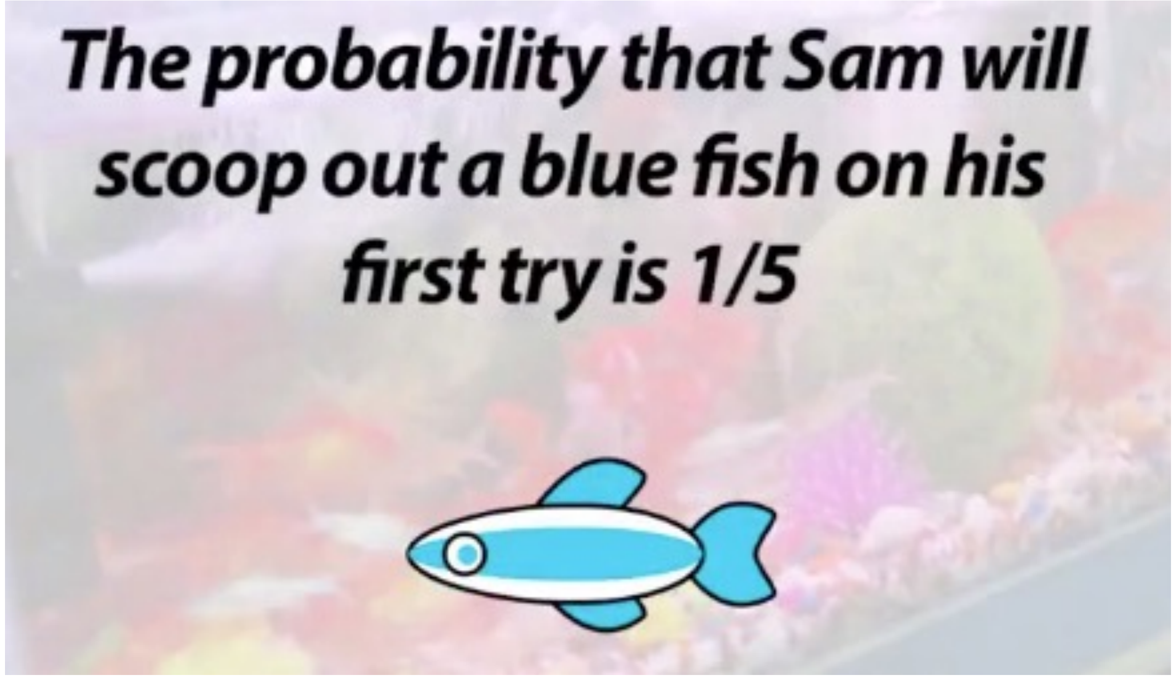
SOLUTION

- In the absence of historical data on meteorites hitting cars, we cannot use the relative frequency approach of Rule 1.
- There are two possible outcomes (hit or no hit), but they are not equally likely, so we cannot use the classical approach of Rule 2.
- That leaves us with Rule 3, whereby we make a subjective estimate.
- In this case, we all know that the probability in question is very, very small. Let's estimate it to be, say, 0.000000000001 (equivalent to 1 in a trillion). *THAT'S SUBJECTIVE ESTIMATE*.

Example of Simple Events



Answer


The probability that Sam will scoop out a blue fish on his first try is 1/5

Example 2

- A committee has 8 female and 12 male members. What is the probability of choosing a female as the president of this committee?

$$\text{Probability of choosing a female as a president} = \frac{\text{number of favourable outcomes } (\text{i.e. females})}{\text{total number of outcomes } (\text{i.e. all members})}$$

- Here the number of females (favourable event) is 8, and the total number of members (outcomes) is 20. Hence:

$$P\,(\text{choosing a female president}) = \frac{8}{20}$$

Example 3

1/3 $\frac{5}{10}$

1/10

- The numbers 1 to 10 are written on separate pieces of paper, folded and put in a box. One number (piece of paper) is drawn from this box.

1.) • What is the probability that this number chosen randomly is 3?

2.) • What is the probability that this randomly chosen number is even?

CALCULATING PROBABILITIES USING SIMPLE EVENTS

Probability of Simple Events

1 - 1/6
2 - 1/6
3 - 1/6
4 - 1/6
5 - 1/6
6 - 1/6

→ 6/6 or 1 or 100%

- Probability of single event occurring.
- A ***simple event*** is an event where all possible outcomes are ***equally likely*** to occur.
- So, the probability of simple events will have all possible outcomes equally likely to happen or occur.

REQUIREMENTS FOR SIMPLE-EVENT PROBABILITIES

1.) • Each probability must lie between 0 and 1.

2) • The sum of the probabilities for all simple events in S equals 1.

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• When it is possible to write down the simple events associated with an experiment and to determine their respective probabilities, we can find the probability of an event A by summing the probabilities for all the simple events contained in the event A.

Definition

- The probability of an event A is equal to the sum of the probabilities of the simple events contained in A.

$S = \{1, 2, 3, 4, 5, 6\}$

$1/6$

Example

- Toss two fair coins and record the outcome. Find the probability of observing exactly one head in the two tosses.

Solution:

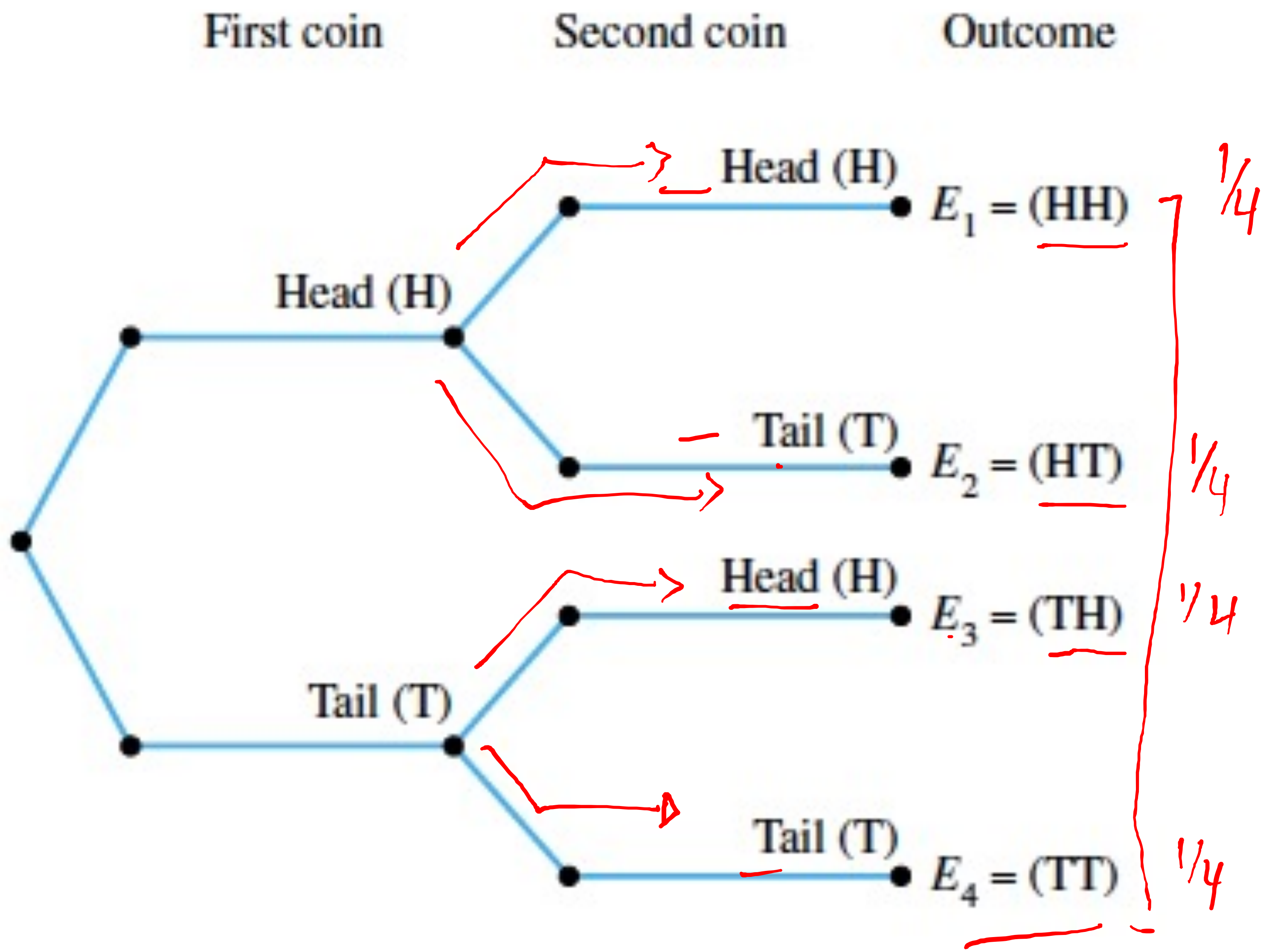
$$P(A) = P(E_2) + P(E_3)$$

$$= \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$= \frac{1}{4} + \frac{1}{4} = \frac{2}{4} \div \frac{2}{2} = \frac{1}{2}$$

Event	First Coin	Second Coin	$P(E_i)$
E_1	H	H	1/4
E_2	H	T	1/4
E_3	T	H	1/4
E_4	T	T	1/4

Example: Using the tree diagram



Example 2:

- The proportions of blood phenotypes A, B, AB, and O in the population of all Caucasians in the United States are reported as .41, .10, .04, and .45, respectively.
- If a single Caucasian is chosen randomly from the population, what is the probability that he or she will have either type A or type AB blood?

Solution to Example 2

0 —— 0.5 —— 1

- The four simple events, A, B, AB, and O, do not have equally likely probabilities. Their probabilities are found using the relative frequency concept as:

$$P(\mathrm{A}) = .41 \qquad P(\mathrm{B}) = .10 \qquad P(\mathrm{AB}) = .04 \qquad P(\mathrm{O}) = .45$$

- The event of interest consists of two simple events, so

$$\begin{aligned} P(\text{person is either type A or type AB}) &= P(\mathrm{A}) + P(\mathrm{AB}) \\ &= .41 + .04 = .45 \end{aligned}$$

Your turn:

First pick	2nd	Simple event	Probability
R_1	R_2	$\boxed{R_1R_2}$	$1/6$
	Y	R_1Y	$1/6$
R_2	R_1	$\boxed{R_2R_1}$	$1/6$
	Y	R_2Y	$1/6$
Y	R_1	YR_1	$1/6$
	R_2	YR_2	$1/6$

- A candy dish contains one yellow and two red candies. You close your eyes, choose two candies one at a time from the dish, and record their colors. What is the probability that both candies are red?

– jar

$$P(\text{both red}) = P(R_1R_2) + (P(R_2R_1))$$

$$\frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$